

Satoshi Tsutsui

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Education

- Ph.D., School of Informatics, Computing, and Engineering, Indiana University, Bloomington, USA. 2021
 - Dissertation: *Rethinking the Role of Training Data for Computer Vision: Scientific Studies of Egocentric Vision*.
 - Advisors: David Crandall (committee chair) and Chen Yu (committee member).
- M.S., School of Informatics, Computing, and Engineering, Indiana University, Bloomington, USA. 2017
 - Advisor: Ying Ding.
- B.E. (with highest honors), Faculty of Science and Technology, Keio University, Tokyo, Japan. 2015

Experience

- Postdoctoral Research Fellow at Nanyang Technological University, Singapore. September 2022 - Present
 - Advisor: Bihan Wen. Founding Lab Director (now Emeritus): Alex Kot.
- Postdoctoral Research Fellow at National University of Singapore. December 2021 - September 2022
 - Advisor: Mike Shou.
- Research Intern at Facebook, USA. December 2020 - June 2021
 - Mentor: Ruta Desai and Karl Ridgeway.
 - Developed visual perception algorithms for AR/VR devices.
- Visiting Ph.D. Student at Fudan University, China. May 2019 - August 2019
 - Mentor: Yanwei Fu.
 - Worked on few-shot visual recognition.
- Visiting Ph.D. Student at Peking University, China. May 2018 - August 2018
 - Mentor: Liangcai Gao.
 - Worked on computer vision for medical images.
- Research Intern at Preferred Networks, Japan. May 2017 - August 2017
 - Mentor: Tommi Kerola and Shunta Saito.
 - Developed semantic segmentation algorithms for autonomous driving.

Publications

1. Satoshi Tsutsui, Winnie Pang, Shuting He, and Bihan Wen. (2026). WBCAtt+: Fine-Grained Pixel-Level Morphological Annotations for White Blood Cell Images. *Medical Image Analysis*. (Impact Factor = 11.8).
2. Patrick Batsell, Satoshi Tsutsui, and Bihan Wen. (2026). Learning Through Little Eyes: Attribute Discrimination Beyond Objects. *IEEE International Conference on Image Processing (ICIP)*.
3. Xueyi Ke, Satoshi Tsutsui, Yayun Zhang, and Bihan Wen. (2025). Discovering Hidden Visual Concepts Beyond Linguistic Input in Infant Learning. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. (Acceptance rate = 2878/13008 = 22.12%).
4. Ziwang Xu, Lanqing Guo, Satoshi Tsutsui, Shuyan Zhang, Alex C Kot, and Bihan Wen. (2025). Digital Staining with Knowledge Distillation: A Unified Framework for Unpaired and Paired-But-Misaligned Data. *IEEE Transactions on Medical Imaging*. (Impact Factor = 10.048).

5. Xiyu Wang, Yufei Wang, [Satoshi Tsutsui](#), Weisi Lin, Bihan Wen, and Alex Kot. (2024). Evolving Storytelling: Benchmarks and Methods for New Character Customization with Diffusion Models. *ACM International Conference on Multimedia (ACMMM)*. (Oral acceptance rate = $174/4385 = 3.97\%$, Overall acceptance rate = $1150/4385 = 26.23\%$).
6. Winnie Pang, Xueyi Ke, [Satoshi Tsutsui](#), and Bihan Wen. (2024). Integrating Clinical Knowledge into Concept Bottleneck Models. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. (Acceptance rate = $858/2771 = 30.96\%$).
7. Juan Hu, Xin Liao, Difei Gao, [Satoshi Tsutsui](#), Qian Wang, Zheng Qin, and Mike Zheng Shou. (2024). Delocate: Detection and Localization for Deepfake Videos with Randomly-Located Tampered Traces. *International Joint Conference on Artificial Intelligence (IJCAI)*.
8. [Satoshi Tsutsui](#), Winnie Pang, and Bihan Wen. (2023). WBCAtt: A White Blood Cell Dataset Annotated with Detailed Morphological Attributes. *Advances in Neural Information Processing Systems (NeurIPS)*. (Acceptance rate = $322/987 = 32.62\%$).
9. Xizhe Xue, Dongdong Yu, Lingqiao Liu, Yu Liu, [Satoshi Tsutsui](#), Ying Li, Zehuan Yuan, Ping Song, and Mike Zheng Shou. (2023). Transformer-based Open-world Instance Segmentation with Cross-task Consistency Regularization. *ACM International Conference on Multimedia (ACMMM)*. (Acceptance rate = $902/3072 = 29.3\%$).
10. Alex Jinpeng Wang, Yixiao Ge, Rui Yan, Yuying Ge, Kevin Qinghong Lin, [Satoshi Tsutsui](#), Xudong Lin, Guanyu Cai, Jianping Wu, Ying Shan, Xiaohu Qie, and Mike Zheng Shou. (2023). All in One: Exploring Unified Video-Language Pre-training. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (Acceptance rate = $2360/9155 = 25.8\%$).
11. [Satoshi Tsutsui](#), Zhengyang Su, and Bihan Wen. (2023). Benchmarking White Blood Cell Classification Under Domain Shift. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*.
12. Zan-Xia Jin, Mike Zheng Shou, Fang Zhou, [Satoshi Tsutsui](#), Jingyan Qin, and Xu-Cheng Yin. (2022). From Token to Word: OCR Token Evolution via Contrastive Learning and Semantic Matching for Text-VQA. *ACM International Conference on Multimedia (ACMMM)*. (Acceptance rate = $690/2473 = 27.9\%$).
13. Eric Zhongcong Xu, Zeyang Song, [Satoshi Tsutsui](#), Chao Feng, Mang Ye, and Mike Zheng Shou. (2022). AVA-AVD: Audio-visual Speaker Diarization in the Wild. *ACM International Conference on Multimedia (ACMMM)*. (Acceptance rate = $690/2473 = 27.9\%$).
14. [Satoshi Tsutsui](#), Xizi Wang, Guangyuan Weng, Yayun Zhang, Chen Yu, and David Crandall. (2022). Action Recognition based on Cross-Situational Action-object Statistics. *International Conference on Development and Learning (ICDL)*.
15. [Satoshi Tsutsui](#), Yanwei Fu, and David Crandall. (2022). Reinforcing Generated Images via Meta-learning for One-Shot Fine-Grained Visual Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)*. (Impact Factor = 24.31).
16. [Satoshi Tsutsui](#), Yanwei Fu, and David Crandall. (2021). Whose hand is this? Person Identification from Egocentric Hand Gestures. *Winter Conference on Applications of Computer Vision (WACV)*. (First round acceptance; Acceptance rate = $496/1241 = 35.4\%$).
17. [Satoshi Tsutsui](#), David Crandall, and Chen Yu. (2021). Reverse-engineer the Distributional Structure of Infant Egocentric Views for Training Generalizable Image Classifiers. *International Workshop on Egocentric Perception, Interaction and Computing (EPIC)*, In conjunction with the *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. (Extended Abstract).
18. [Satoshi Tsutsui](#), Ruta Desai, and Karl Ridgeway. (2021). How You Move Your Head Tells What You Do: Self-supervised Video Representation Learning with Egocentric Cameras and IMU Sensors. *International Workshop on Egocentric Perception, Interaction and Computing (EPIC)*, In conjunction with the *IEEE International Conference on Computer Vision (ICCV)*. (Extended Abstract).
19. [Satoshi Tsutsui](#), Arjun Chandrasekaran, Md Reza, David Crandall, and Chen Yu. (2020). A Computational Model of Early Word Learning from the Infant’s Point of View. *Annual Conference of the Cognitive Science Society (CogSci)*. (Oral Acceptance Rate = $177/811 = 22\%$).
20. [Satoshi Tsutsui](#), Yanwei Fu, and David Crandall. (2019). Meta-Reinforced Synthetic Data for One-Shot Fine-Grained Visual Recognition. *Advances in Neural Information Processing Systems (NeurIPS)*. (Poster Acceptance Rate = $1428/6743 = 21\%$).

21. Satoshi Tsutsui, Dian Zhi, Md Alimoor Reza, David Crandall, and Chen Yu. (2019). Active Object Manipulation Facilitates Visual Object Learning: An Egocentric Vision Study. *International Workshop on Egocentric Perception, Interaction and Computing (EPIC)*, In conjunction with the *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. (Extended Abstract).
22. Zheng Gao, Gang Fu, Chunping Ouyang, Satoshi Tsutsui, Xiaozhong Liu, Jeremy Yang, Christopher Gessner, Brian Foote, David Wild, and Ying Ding. (2019). edge2vec: Representation Learning Using Edge Semantics for Biomedical Knowledge Discovery. *BMC Bioinformatics*. 20(1), 306. (Impact Factor = 3.213).
23. Satoshi Tsutsui, Tommi Kerola, Shunta Saito, and David J Crandall. (The first three authors have equal contribution). (2018). Minimizing Supervision for Free-space Segmentation. *Workshop on Autonomous Driving (WAD)*, In conjunction with the *Conference on Computer Vision and Pattern Recognition (CVPR)*.
24. Satoshi Tsutsui, Sven Bambach, David Crandall, and Chen Yu. (2018). Estimating head motion from egocentric vision. *ACM International Conference on Multimodal Interaction (ICMI)*.
25. Ting-Ting Liang, Mengyan Sun, Liangcai Gao, Jing-Jing Lu, and Satoshi Tsutsui. (2018). APNet: semantic segmentation for pelvic MR image. *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*.
26. Satoshi Tsutsui, Zheng Gao, Yuzhuo Wang, Guilin Meng, and Ying Ding. (2018). A case study on viziometrics: What's the role of western blots in Alzheimer's Disease literature?. *iConference*. (Poster).
27. Satoshi Tsutsui, and David J Crandall. (2017). A data driven approach for compound figure separation using convolutional neural networks. *IAPR International Conference on Document Analysis and Recognition (ICDAR)*. (Oral Acceptance Rate = $52/409 = 13\%$).
28. Satoshi Tsutsui, Tommi Kerola, and Shunta Saito. (2017). Distantly supervised road segmentation. *Workshop on Computer Vision for Road Scene Understanding and Autonomous Driving (CVRSUAD)*, In conjunction with the *IEEE International Conference on Computer Vision (ICCV)*.
29. Baitong Chen, Satoshi Tsutsui, Ying Ding, and Feicheng Ma. (2017). Understanding the topic evolution in a scientific domain: An exploratory study for the field of information retrieval. *Journal of Informetrics*. 11(4), 1175-1189. (Impact Factor = 3.879).
30. Satoshi Tsutsui, Guilin Meng, Xiaohui Yao, David Crandall, and Ying Ding. (2017). Analyzing Figures of Brain Images from Alzheimer's Disease Papers. *iConference*. (Poster).
31. Satoshi Tsutsui, and David Crandall. (2017). Using artificial tokens to control languages for multilingual image caption generation. *Language and Vision Workshop*, In conjunction with the *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. (Extended Abstract).
32. Satoshi Tsutsui, Yi Bu, and Ying Ding. (2017). Using machine reading to understand Alzheimer's and related diseases from the literature. *Journal of Data and Information Science*. 2(4), 81-94. (Impact Factor = 1.771).
33. Satoshi Tsutsui, Ying Ding, and Guilin Meng. (2016). Machine reading approach to understand Alzheimer's disease literature. *International Workshop on Data and Text Mining in Biomedical Informatics (DTMBIO)*, In conjunction with the *ACM Conference on Information and Knowledge Management (CIKM)*.

Awards

- NeurIPS student travel award. 2019
- ICDAR student travel award. 2017
- Scholarship for Study Abroad from Yoshida Scholarship Foundation. 2015 - 2017
- Keio University Scholarship for Excellent Undergraduate Students. 2012